IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

SOFTWARE SYSTEMS

Friday, 9 May 2025, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : J. Pitt

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Final Exam
ELEC50014 - Software Systems

Time: 2 Hours
Total Marks: 60

Question 1

(a) [**10 marks**] Draw an Entity Relationship diagram for the following description of a Research Conference Management System. Show the entities, relationships, cardinalities of relationships, participation types of relationships, attributes, and primary keys for a system based on this description.

A university is developing a Research Conference Management System to handle conference submissions and reviews.

Each researcher has a unique researcher ID, a name (first and last), an email, and multiple affiliations (names of institutions they belong to).

A conference is identified by a unique conference ID, has a name, and a set of tracks.

Each track has a unique title and an assigned track chair (who is also a researcher). No researcher may chair more than one track. A track may be a sub-track of another track within the same conference. Each sub-track has at most one parent track, while a parent track can have multiple sub-tracks.

Researchers submit papers to conferences. A researcher may submit one or more papers to one or more conferences, and each submission has a date. A paper has a unique ID, a title, an abstract, and possibly multiple authors (who are also researchers). There may be researchers in the system who have not submitted or authored any papers. However, each paper in the system must have been submitted.

A paper may have any number of associated revisions. A paper can undergo multiple revisions, each with a revision number (unique for that paper), a date, and a file link.

Each paper is assigned one or more reviewers (also researchers) to conduct a review of that paper. Each review has a unique ID, reviewer comments, and rating. A paper may get reviewed more than once.

(b) [**5 marks**] Convert the Entity Relationship model from (a) into a relational schema. Highlight the primary key of each relation in the schema.

(c) [10 marks (5 + 5)] The following relational schema represents a part of a hospital's appointments management system.

Doctors(doctorID, name, speciality)
Patients(patientID, name, dateOfBirth)
Appointments(appointmentID, patientID, doctorID, appointmentDate)

Write SQL queries to answer the following questions.

(i) Count the number of appointments for a given doctor (identified by doctor_name) between two specified dates (d1 and d2).

(ii) Find the names of all patients who have seen a doctor specializing in 'Cardiology'.

Question 2

(a) [6 marks (3 + 3)] Answer the following questions **briefly**.

(i) Why is iterative DNS resolution generally preferred over recursive DNS resolution when querying an authoritative server?

(ii) HTTP is called a stateless protocol, even though it uses TCP to maintain connections between a web client and server. Explain why this is the case.

(b) [8 marks] Design a stop-and-wait-based protocol for reliable data transmission in a two-hop network where a packet must be transferred from node A to B to C. The sender A will not send a new packet until it knows that B has received the current packet correctly and successfully forwarded it to C. The channels are corruption-free but have variable delay and may lose packets during transmission.

Draw state machines (one each) for A and B. It is sufficient to show the working of the protocol for sequence number 0, as the case for sequence number 1 is symmetric. You may use the functions listed in Table 1 (on Page 3) to indicate actions and events in the state transition labels. Moreover, Figure 1 shows the state machine at C (for sequence number 0 only). Notice that C does not use the timer as it does not send out data packets.

(c) [6 marks] Host A is transmitting three packets simultaneously to Host B through an intervening router R. Each packet has a length of l bits. Assume that there are no other packets on the network and router R has sufficient queueing capacity. It transmits a packet on its outgoing link as soon as the previous packet has been transmitted. The transmission rate on A's outgoing link is r_A bps, while the transmission rate on R's outgoing link is r_R bps. The length of the link A–R is d_1 meters, the length of the link R–B is d_2 meters, and the propagation speed on both links is v meters per second. We assume that $r_R < r_A$, which results in all three packets being available for transmission at R at the same time. Express the average end-to-end delay for the transmission of three packets from A to B in terms of l, r_A, r_R, d_1, d_2, and v.

Function	Description
`data_received()`	indicates data received from the upper layer
`data_delivered()`	delivers data to the upper layer
`has_seq(pk, seq)`	indicates that data packet `pk` has sequence number `seq`
`pk = create_packet(seq)`	creates data packet `pk` with sequence number `seq`
`send_to(node, pk)`	sends packet or acknowledgment `pk` to the specified node (A, B, or C)
`pk = receive_from(node)`	indicates reception of packet `pk` (data or acknowledgment) from the specified node
`timeout()`	indicates timer has expired
`start_timer()`	starts the timer
`stop_timer()`	stops the timer
`pk = create_ack(seq)`	creates acknowledgment packet `pk` with sequence number `seq`
`is_ack(pk, seq)`	indicates that `pk` is an acknowledgment with sequence number `seq`

Table 1: Functions for the protocol in Question 2 (b)

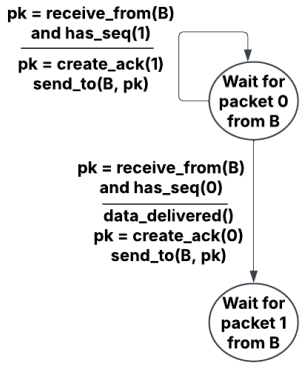


Figure 1: State machine at node C (for sequence number 0 only)

Question 3

(a) [**5 marks**] A router interconnects three subnets, referred to as Subnet1, Subnet2, and Subnet3, where all interfaces must have the prefix `192.168.16.0/21`. Subnet1 needs to support 1010 interfaces, while Subnet2 and Subnet3 each require 400 interfaces. Determine a set of network addresses in CIDR notation that satisfy these constraints, and specify the subnet mask and IP address range for each subnet.

(b) [**4 marks**] A network consists of n routers. A new router R is added as a source node, meaning it will not be a destination for any other router but may transmit to any other router

in the network. Router R is connected to the network via k existing routers, which are its neighbors. Each of these neighbors has a fully updated routing table. To compute its own routing table, router R obtains a copy of the routing tables from its k neighbors and uses them to compute the shortest paths to all other routers. Express the time complexity of this computation in terms of n and k in the following cases:

- The network is using link-state routing based on Dijkstra's shortest path algorithm.
- The network is using distance-vector routing based on Bellman-Ford algorithm.

(c) [**6 marks (3 + 3)**] Answer the following questions **briefly**.

(i) Both ARP and DHCP protocols generate broadcast traffic. Explain why this is required.

(ii) A gateway router implements NAT functionality using an appropriate data structure. For each outgoing and incoming packet, the data structure is accessed using insert and lookup functions, respectively. Specify the input parameters and output values for the insert and lookup functions. Explain why these inputs and outputs are necessary.